

MATERIAL SAFETY DATA SHEET

Total Solutions
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GENERAL INFORMATION NUMBER: (414) 354-6417
CHEMTREC: (800) 424-9300

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I - Product Identification

MR²

PRODUCT CODE: 0210

CHEMICAL FORMULATION: Powdered glass cleaner.

NFPA HAZARD IDENTIFICATION SYSTEM:

HEALTH: 2

FLAMMABILITY: 0

REACTIVITY: 0

HAZARD RATING:

4 - Extreme;

3 - High;

2 - Moderate;

1 - Slight;

0 - Insignificant

II - Hazardous Ingredients

Values reported as TWA unless noted.

<u>SUBSTANCE</u>	<u>APPROX</u>	<u>OSHA</u>	<u>ACGIH</u>	<u>EPA 40 CFR:</u>			<u>CAS #</u>
	<u>%</u>	<u>PEL</u>	<u>TLV</u>	<u>302</u>	<u>355</u>	<u>372</u>	
Sodium Sulfate	30.0-40.0	N/E	N/E	N	N	N	7757-82-6
Sulfamic acid	15.0-25.0	N/E	N/E	N	N	N	5329-14-6
Crystalline silica (respirable fraction)	15.0-25.0	(0.1 mg/m ³)	(0.1 mg/m ³)	N	N	N	14808-60-7
Oxalic acid	10.0-20.0	1 mg/m ³	1 mg/m ³	N	N	N	144-62-7
Diatomaceous Earth	1.00-10.0	6 mg/m ³	10 mg/m ³	N	N	N	68855-54-9

Key: **PEL:** Permissible Exposure Limit **TLV:** Threshold Limit Value **C:** Ceiling level **STEL:** Short Term Exposure Limit
N/A: Not Applicable **N/D:** Not Determined **N/E:** Not Established **Y:** Yes **N:** No
302: CERCLA List of Hazardous Substances and Reportable Quantities (40 CFR 302.4).
355: SARA TITLE III / List of Extremely Hazardous Substances for Emergency Planning and Notification (40 CFR 355).
372: SARA TITLE III / List of Toxic Chemicals subject to Release Reporting (Community Right to Know) (40 CFR 372).

III - Physical Data

BOILING POINT (°F): N/A

SPECIFIC GRAVITY (WATER = 1): N/A

VAPOR PRESSURE (mm Hg): N/A

VOC CONTENT (% by weight): N/A

VAPOR DENSITY (AIR = 1): N/A

EVAPORATION RATE (WATER = 1): N/A

SOLUBILITY IN WATER: Complete

pH: N/D

APPEARANCE AND ODOR: Free flowing off white powder, no odor.

IV - Fire and Explosion Hazard Data

FLASH POINT (°F): N/A

(TEST METHOD): Closed cup

FLAMMABLE LIMITS IN AIR (VOLUME %)

UPPER: N/A

LOWER: N/A

EXTINGUISHING MEDIA: Water, foam, carbon dioxide, dry chemical.

SPECIAL FIRE FIGHTING PROCEDURES: Cool fire exposed containers with water fog. Firefighters should be equipped with full protective gear including self-contained breathing apparatus.

UNUSUAL FIRE AND EXPLOSION HAZARD: Foam or water may cause frothing. Emits highly toxic and irritating fumes in a fire. Closed containers exposed to heat may explode.

V - Reactivity Data

STABILITY: Stable

INCOMPATIBILITY: Strong oxidizers and alkali, nitrates, nitrites, carbonates, sulfides, cyanides, sodium oxide, sulfur oxides.

CONDITIONS TO AVOID: Extreme heat & aluminum.

HAZARDOUS DECOMPOSITION PRODUCTS: Thermal decomposition may produce oxides of sulfur, nitrogen, ammonia & formic acid.

HAZARDOUS POLYMERIZATION: Will not occur.

CONDITIONS TO AVOID: None

VI - Health Hazard Data

ROUTES OF ENTRY **INHALATION:** X **EYE CONTACT:** X **SKIN CONTACT:** X **INGESTION:** X
INGREDIENTS THAT ARE CONSIDERED BY OSHA, NTP, IARC TO BE SUSPECTED HUMAN CARCINOGENS: Crystalline silica (CAS #14808-60-7)

EFFECTS OF OVEREXPOSURE

IF IN EYES: Severe irritation & burns. Permanent damage may result.

IF ON SKIN: Moderate to severe irritation. Prolonged skin contact may cause burns.

IF SWALLOWED: Nausea, vomiting. May be harmful if swallowed.

IF INHALED: Extremely harmful to mucous membranes and upper respiratory tract. Respirable crystalline silica can cause silicosis.

EMERGENCY AND FIRST AID PROCEDURES

IF IN EYES: Flush eyes and under eyelids with plenty of cool water for at least 15 minutes. If irritation persists, obtain medical attention.

IF ON SKIN: Flush with water for 15 minutes while removing contaminated clothing. Launder clothing separately before reuse. If irritation persists, obtain medical attention.

IF SWALLOWED: Contact physician or poison control center immediately. Give affected person 1 to 2 glasses of water. Do not induce vomiting unless instructed by a physician or poison control center. Never give anything by mouth to an unconscious person.

IF INHALED: Remove person to fresh air. If breathing has stopped, administer artificial respiration. If breathing is difficult administer oxygen.

VII - Spill or Leak Protection

STEPS TO BE TAKEN IF MATERIAL IS RELEASED OR SPILLED: Ventilate area and remove all sources of ignition. Scoop up spilled material, avoiding generation of dust, and place in a properly marked closed container for proper disposal.

WASTE DISPOSAL METHOD: Consult local environmental authorities.

VIII - Special Protection Information

RESPIRATORY PROTECTION: Use with adequate ventilation. Do not breathe dust. If recommended Exposure Limits are exceeded wear a NIOSH approved respirator, following manufacturer's recommendations.

VENTILATION

LOCAL: Recommended

MECHANICAL: Not required

PROTECTIVE GLOVES: Chemical resistant.

EYE PROTECTION: Safety glasses or goggles.

OTHER PROTECTIVE EQUIPMENT: Eye wash station.

PRECAUTIONS TO BE TAKEN IN HANDLING AND STORAGE: Store in a cool, dry place away from heat or open flame. Keep container tightly closed when not in use.

OTHER PRECAUTIONS: Keep out of reach of children.

IX - Transportation Information (ground transportation only)

DOT PROPER SHIPPING NAME: Consumer Commodity

DOT CLASS: ORM-D

DOT ID NUMBER: None

DOT PACKING GROUP: None

The shipping information listed above applies only to non-bulk (< 119 gallons) containers of this product. This product may have more than one proper shipping name depending on packaging, product properties, & mode of shipment. If any alteration of packaging, product, or mode of transportation is further intended, different shipping names and labeling may apply.

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